PREDICTING STUDENTS' DROPOUT AND ACADEMIC SUCCESS
ICT 110 - RxPulseMH
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DATASET DESCRIPTION

The project will use the Predict Students' Dropout and Academic Success dataset from the UCI Machine Learning Repository (Dataset ID: 697). The UCI repository lists 4,424 student records, 36 predictor features, and one target variable. Each record represents an individual student [1]. The dataset contains demographic, socioeconomic, admission, macroeconomic, and academic performance information from the first and second semesters. It was collected from the Polytechnic Institute of Portalegre in Portugal, covering academic years 2008/2009 to 2018/2019. The data came from several sources, including the Academic Management System (AMS), Support System for the Teaching Activity (PAE), Directorate-General for Higher Education (DGES), and Contemporary Portugal Database (PORDATA). It is publicly available under the Creative Commons Attribution 4.0 International (CC BY 4.0) license [1], [2].

Dataset Link:
https://archive.ics.uci.edu/dataset/697/predict%20students%20dropout%20and%20academic%20success

PROBLEM STATEMENT

Student dropout negatively impacts institutional success and graduation rates. Identifying at-risk students allows academic administrators and advisers to provide timely, targeted support. This project utilizes **supervised binary classification** to predict whether a student will ultimately **graduate or drop out**. By leveraging a comprehensive timeline of demographic, socioeconomic, and academic performance features across two semesters, the model aims to accurately classify eventual academic outcomes and assist in the university's retention efforts.

CHOSEN METRIC

The primary evaluation metric for this project is the **Macro-Averaged F1-Score (Macro-F1)**. After focusing the dataset exclusively on two outcomes, the classes remain imbalanced (approximately 61% Graduate and 39% Dropout). Standard accuracy can be misleading in this scenario, as it naturally favors the majority class. Macro-F1 addresses this by calculating the F1-score for both classes independently and taking the average. This ensures the model is evaluated fairly and must perform well in identifying both dropouts and graduates without majority-class bias.

BASELINE IDEA

To evaluate model performance, this project will establish a two-part baseline. First, a Majority Class (Dummy) Classifier will be used; by predicting the most common outcome ("Graduate") for every student, it sets a minimum floor and demonstrates the limitations of standard accuracy. Second, an untuned Logistic Regression model will serve as a basic mathematical baseline. To be considered successful, our final optimized machine learning models must significantly outperform these baselines in their Macro-F1 score, proving they have learned meaningful academic and demographic patterns rather than relying on simple guessing.

According to the original research paper, all data are anonymized, and compliance with the institution's Privacy and Personal Data Processing Policy is ensured in accordance with the General Data Protection Regulation (GDPR) [2]. However, demographic and socioeconomic factors may contribute to unequal model performance across student groups. Since the dataset comes from a single higher education institution and covers academic years 2008/2009 to 2018/2019, its findings may not fully represent students from other institutions or current academic populations. The model will be used solely for academic analysis and as supplementary decision support information to help identify students who may need academic assistance, not as the sole basis for decisions or to restrict educational opportunities.

REFERENCES

[1] V. Realinho, M. Vieira Martins, J. Machado, and L. Baptista, "Predict Students' Dropout and Academic Success," UCI Machine Learning Repository, 2021, doi: 10.24432/C5MC89.
[2] V. Realinho, J. Machado, L. Baptista, and M. V. Martins, "Predicting Student Dropout and Academic Success," Data, vol. 7, no. 11, Art. no. 146, 2022, doi: 10.3390/data7110146.